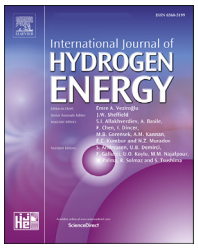
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Review Article

A systematic review of hybrid renewable energy systems with hydrogen storage: Sizing, optimization, and energy management strategy



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HIGHLIGHTS

- An efficient energy storage system is essential for managing intermittent energy supply.
- A fuel cell generates electricity by combining hydrogen and oxygen to form water.
- Soft computing techniques use both single and hybrid algorithms.
- Deterministic rule-based controllers operate based on pre-defined rules.
- Energy Management System coordinate the flow of power between different components.

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ABSTRACT

Renewable energy systems (RESs) have played a vital role in meeting the continuously fast-rising energy demand. While RESs provide limitless and ecologically good possibilities, putting RESs in place has proven to be a challenging task due to their intermittent energy supply. Battery energy storage systems have widely been used in microgrids to manage the issue of intermittent energy supply caused by the RESs. The disadvantages of batteries include, large size, limited lifespan, and high cost. For this reasons, energy planners are looking into hydrogen-based storage systems as a potential solution. This paper aims to provide the recent advancements in the sizing and energy management of hybrid RES (HRES), as well as conventional methods. To achieve this, the authors have employed a review strategy that involves several specific tasks, such as preparing a database of articles related to Hybrid RESs with hydrogen storage, designing inclusion/exclusion criteria to filter out relevant articles, and analyzing each article from critical viewpoints. Thus, this paper elaborates on the general formulation framework for optimization, the classification and review of different optimization methods, and the literature on the application and

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research of Energy Management Systems (EMS). Additionally, the review presents an analysis and comparison of various system optimization methods and EMS through a study of several cases from the literature. It is hoped that the review will act as a basic framework for researchers looking to investigate HRES incorporated with hydrogen storage.

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Nomenclature

ACO	Ant colony optimization
ANN	Artificial neural network
BBO	Biogeography based optimization
BES	Battery energy storage
BFA	Bacteria food algorithm
CFR	Capital recovery factor
CHP	Combine heat power
CS	Cuckoo search
DE	Differential evolution
DG	Distributed generator
HRES	Hybrid renewable system
IFFA	Improved fruit fly algorithm
LIP	Load interruption probability
LPS	Loss of power supply
MPC	Model predictive control
NSGAII	Non-dominated sorting GA
RES	Renewable energy source
SoC	State of charge
SSR	Self-sufficiency ratio
TAC	Total annual cost
TE	Total energy
TLCC	Total life cycle cost
TNPC	Total net present cost
A_{PV}	Area of photovoltaic
A_{wt}	Area of wind turbine
E_{CO_2}	Carbon dioxide emission
E_{SO_2}	Sulphur oxide emission
E_{wt}	Wind turbine energy
E_{Load}	Load energy
E_{serv}	Energy served
In_C	Inverter cost
N_{bat}	Number of batteries
N_{sol}	Number of solar
N_{wt}	Number of wind turbine
E_{surp}	Surplus power
C_{inv}	Cost of inverter
C_{sol}	Cost of solar
C_{wt}	Cost of wind turbine
P_{gen}	Power generated
P_{inv}	Inverter power
P_L	Load power
T_{npc}	Total net present cost
P_{PV}	PV power
P_{wt}	Wind turbine power

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Introduction

Hydrogen is a colourless, odourless, and highly flammable gas. The lightest and most abundant element in the universe, hydrogen can be produced from various sources, including water, natural gas, and biomass [1–4]. When hydrogen is used as a fuel, the gas can be converted to electricity in a hydrogen fuel cell (FC). Hydrogen fuel cells are highly efficient, with an efficiency of up to 60%, and produce no harmful emissions since the only by-products are water and heat [5,6] with a long lifespan, operate silently, and require minimal maintenance. However, the widespread adoption of hydrogen fuel cells as a source of energy is still limited due to high production costs and infrastructure challenges. Hydrogen is an environmentally acceptable fuel that does not emit any molecules that are harmful to the environment when burned or oxidised at low temperatures [7,8]. Currently, hydrogen is classified into several categories based on the production, including green, blue, grey, brown, and black. Green hydrogen is made entirely from renewable sources and has no carbon emissions. Blue hydrogen is produced with CO_2 capture and sequestration, while grey hydrogen is produced without CO_2 capture. Brown hydrogen is made from brown coal gasification, and black hydrogen is made from black coal gasification. Experts predict that the cost of green hydrogen will drop below $1 per kg by 2050, making green hydrogen more competitive. This stresses the need of continued research and development in the field of hydrogen energy, as hydrogen is predicted to become the preferred fuel of choice in the future due to the high energy content and ecologically beneficial features [9].

Hydrogen fuel cell technology has received a lot of attention because of the high efficiency and low levels of noise and pollution in comparison to other chemical energy conversion and consumption systems. The proton exchange membrane fuel cell (PEMFC) is the best form of fuel cell for portable power devices and distributed power production systems. This is due to the fact that it runs at lower temperatures and may start and stop faster than other types of fuel cells [10–12]. According to [13], hydrogen (H_2) can be considered a form of renewable energy storage because of the ability to be produced through the process of electrolysis and stored in tanks. Excess energy can be stored in a H_2 tank by using an electrolyzer, which can later be used to make up for any shortage in generation by providing energy to the fuel cell (FC). Adding FC to a hybrid renewable energy system (RES) can reduce the size of the BES, extend lifespan, and enhance overall system performance [14]. Therefore, after the system configuration is established for renewable energy (RE) systems, either standalone or grid-connected, the primary challenge is the sizing of the system components and the development of an effective energy management strategy (EMS) to achieve the desired goals without degrading the system performance [15]. In other words, the key determinants of both the start-up and operating expenses of the RE system are component size and EMS. Fig. 1 displays the graphical abstract for the article.

The aim of this review article is to update readers on the latest advancements in the field of optimization and EMS of hybrid renewable energy systems integrated with hydrogen technologies. This article also serves as a guide for future research in this area, as well as providing a tool for readers to select the most suitable combination of renewable energy sources along with hydrogen storage for a specific location to meet the energy demand.

Contribution of this review paper

Earlier research studies have played an important role in improving the knowledge of researchers on the optimal sizing and energy management strategy (EMS) of hybrid renewable energy systems (HRESs). Researchers have primarily focused on examining optimization methods for HRES that use photovoltaic (PV) and wind turbines (WT) along with other energy sources such as diesel generators. However, there has been a lack of review articles that discuss the energy management and optimization techniques of hydrogen (H_2) storage based HRES that incorporate but not limited to biomass, micro-hydro, wind, photovoltaic, biofuels, geothermal, ocean energy, and others. To address this gap, this article provides a systematic analysis of various energy management and optimization techniques, including their economic, technical, environmental, and social impacts, that have been employed by the research community over the past decade.

The review also provided essential formulations for assessing the cost of energy (COE), net present cost (NPC), Emission CO_2, and reliability indices in sizing HRES, given that each metric indicates a particular aspect of the system performance and has a distinctive purpose. This will help energy system designers and investors to understand the strengths and weaknesses of the system, thereby enabling decision makers to make informed decisions about the HRES design and implementation.

Systematic literature review

To conduct a thorough review of existing literature on system optimization and Energy Management Systems (EMS) of a HRES integrated with hydrogen energy storage, the following steps were followed. The first step in the process involved gathering numerous research papers from various research

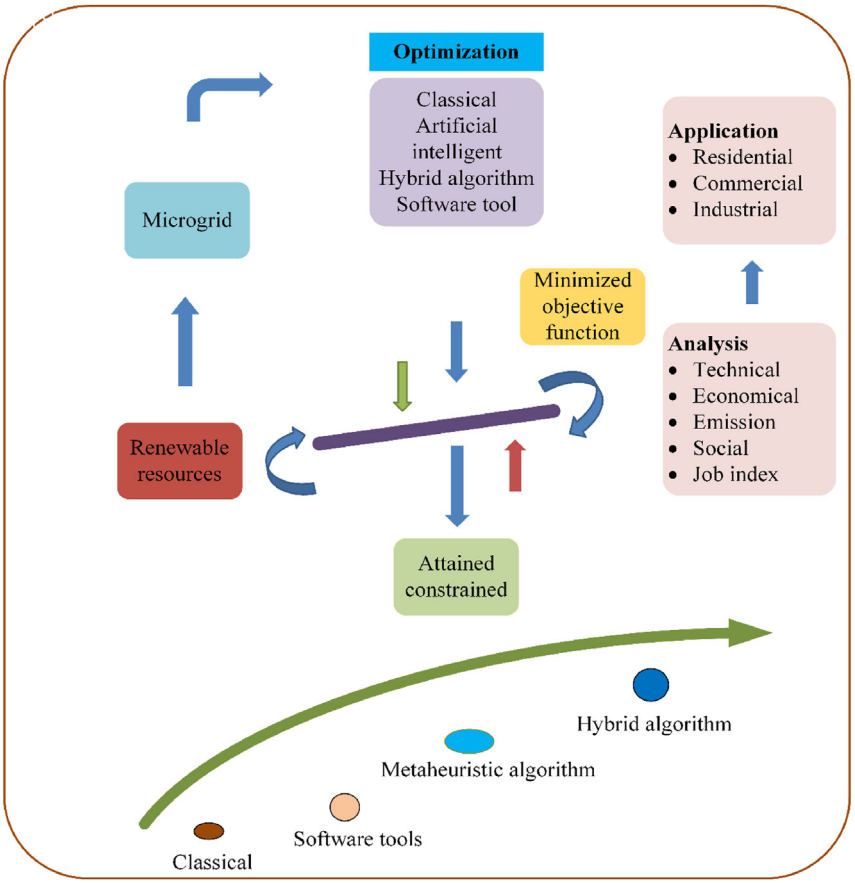


Fig. 1 – Graphical abstract of the article.

databases and websites, including IEEE digital explorer, Science Direct, Springer, Scopus, and Web of Science. The authors searched for these papers using a range of relevant keywords related to the topic. Some of the keywords used in the search included "hydrogen storage", "hydrogen fuel cell", "electrolyser", "optimization software tools", "optimal sizing of HRES with hydrogen storage technology", "energy management and optimization of HRES ″, "techno-economic analysis of HRES", "sizing of standalone microgrid system", "heuristic and metaheuristic energy management optimization", "HRES optimization based on analytical technique", and all possible combinations of these terms.

After gathering the research papers, the authors proceeded to screen the papers using predetermined inclusion and exclusion criteria. The inclusion criteria included studies that focused on hybrid renewable energy systems integrated with hydrogen energy storage, optimization techniques, and EMS. Exclusion criteria included studies that focused on single-source renewable energy systems, studies that did not use optimization techniques. Finally, the authors synthesized the findings from the selected studies to present a comprehensive review of the current state of the art in system optimization and EMS of hybrid renewable energy systems integrated with hydrogen energy storage. Areas that require additional study and development to increase the performance and efficiency of such systems were identified. The authors were inspired by the strategies described in references [16,17].

There was no specific time frame set for the literature search. Initially, the search resulted in 746 articles from the Science Direct, Scopus, and Web of Science (WoS) database, as shown in Fig. 2. These results were then combined, and duplicates were eliminated using Endnote software. The articles were then filtered to include only journal articles and conference papers, published in English language. The abstract and conclusion sections of the remaining articles were examined to assess their relevance to the purposes of this review study. This narrowed down the number of articles to 157. Additionally, a cross-reference check was conducted, which resulted in the identification of 19 more articles. To summarise, each gathered article is read with the goal of gaining a thorough understanding of research findings.

Hydrogen energy storage system (HESS)

Hydrogen plays a vital role in promoting the development of renewable energy applications such as wind and solar power generation [18,19]. The exceptional properties of hydrogen have made hydrogen a popular contender for a fuel source that does not contribute to carbon emissions in the future.

As the world energy needs continue to grow, hydrogen is predicted to be a vital component of upcoming energy infrastructure. Additionally, hydrogen is being utilized as a means of transferring energy with the aim of reducing society's

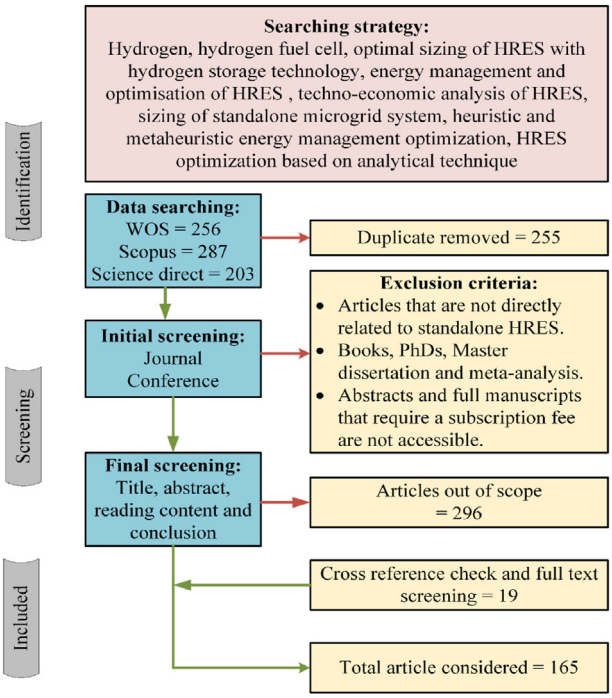


Fig. 2 – PRISMA model of the paper selection criteria.

reliance on fossil fuels that release carbon and mitigating their harmful environmental impact [20,21]. The employment of hydrogen-based energy systems seems to be a highly viable remedy with the potential to make substantial contributions towards environmental improvement and sustainability. Hydrogen has demonstrated potential for serving as a fuel for vehicles, a means of electricity storage through fuel cells, and various other beneficial attributes in the fields of chemistry and metallurgy [22,23]. The availability of multiple possible sources is a crucial factor contributing to the attractiveness of hydrogen as a carrier of energy. Hydrogen can be utilized in large, centralized facilities, semi-central facilities of medium size, or small distributed units placed directly at or near the point of consumption [24,25]. Hydrogen generated through renewable energy sources is expected to fulfil or surpass the energy storage demands of renewable energy systems [26–29]. Ongoing initiatives in the US, Canada, China, Germany, and Japan, provide evidence of the efficacy of such undertakings [30,31]. These projects demonstrate that a significant amount of hydrogen can be produced and stored for later use in a wide range of industrial operations. Furthermore, fuel cells are utilized to generate electricity using hydrogen [32]. Although hydrogen can be generated using various industrial methods, such as the reforming process from chemical compounds, producing hydrogen through renewable energy sources can help reduce the impact of climate change and environmental pollution. The production of hydrogen using electricity generated from traditional power sources like nuclear power stations and thermal power stations remains costly. Therefore, renewable energy sources offer an optimistic solution for producing green hydrogen [33–35].

The hydrogen energy storage system can initially be viewed as an ideal storage system that takes in hydrogen during the charging process, stores for an extended period without any loss, and releases it during the discharging process based on the current energy demand. These three activities do not significantly rely on pressure, heat, or dynamics [36]. Furthermore, hydrogen energy storage systems have a longer lifespan of approximately 25 years when compared to lithium-ion batteries. Over this time, there is no decline in the performance of the hydrogen energy storage system, and the properties do not significantly deteriorate.

A fuel cell generates electricity by combining hydrogen and oxygen to form water. The core components of a fuel cell are electrodes and an electrolyte, with the specific type of electrolyte determining the type of fuel cell. To achieve significant voltage output, multiple fuel cells are combined to form a stack [37]. The proton exchange membrane fuel cell (PEMFC) is the most commonly used type of fuel cell in hydrogen-based hybrid renewable energy systems (HRES). Several research studies have suggested numerous wind or HRES that incorporate hydrogen storage, some of which have already been employed in the HRES. In wind-based energy systems, using hydrogen energy storage system (HESS) has proven to be one of the most effective solutions for mitigating rapid fluctuations in wind power, as a result of the numerous advantages such as higher energy density and lower volume compared to gasoline.

Optimization techniques applied for sizing of HRES with hydrogen storage

Researchers have attempted to optimize HRES-H_2 using various techniques such as classical optimization, soft computing techniques, and hybrid metaheuristic optimization. While classical methods have limitations such as inflexibility and long calculation time, soft computing techniques like PSO, GA, and HS have been used to investigate the combination of solar and wind power with hydrogen technologies. In their research [68], utilized PSO to optimize a photovoltaic-hydrogen system and minimize the cost of energy (COE) and CO_2 emissions. In the paper [38], the Multi-Objective Modified Firefly Algorithm (MOMFA) is utilized to optimize the size and operational strategy of a PV-hydrogen/battery storage system. In the meantime [39], proposed a two-stage, multi-objective EMS based on FLCs for a microgrid that includes both battery and hydrogen storage systems with a view to optimize the techno-economic aspects of the MG by using the newly developed EMS. In [40], the authors presented two EMS algorithms that were tested on a microgrid testbed. The first algorithm's objective was to preserve the balance between load and generation with the assistance of a battery bank while decreasing the use of diesel generators, while the second algorithm aimed to minimize the total operation costs of the microgrid. Python was used to develop both algorithms, along with several open-source libraries for optimization, communication, and GUI design. A multi-period energy model based on P-graphs, utilizing hydrogen as an energy storage medium to meet the variable electrical and thermal energy requirements of an island is proposed by the authors in [41]. The paper [42] introduces an optimization framework based on mixed-integer linear programming (MILP) to evaluate the energy system of the off-grid village of Ginostra, located on

Table 1 – Summary of optimization techniques applied on HRES with hydrogen storage (H_2).

Ref.	System configuration						Techniques	Objective of the study
	PV	WT	Bio	D_{gen}	BES	HESS		
[49]	✓	✓	✗	✗	✓	✓	Homer	Cost (TNPC)
[50]	✓	✗	✗	✗	✗	✓	Homer	Minimization of cost (LCOE) and emission
[51]	✓	✗	✗	✗	✗	✓	ABC-PSO	Reliability and cost
[52]	✓	✓	✗	✗	✗	✓	Homer	Minimization of cost (LCOE)
[53]	✓	✓	✗	✗	✗	✓	PSO	Cost (NPV), and reliability (SSR)
[54]	✓	✓	✗	✓	✗	✓	MILP	Minimization of cost (COE) and emission (CO_2)
[55]	✓	✗	✗	✓	✓	✓	MILP	Minimization of cost (COE) and emission (CO_2)
[56]	✓	✗	✗	✗	✗	✓	MILP	Minimization of cost (ASC) Reliability (LPSP)
[57]	✓	✗	✗	✗	✓	✓	MILP	Minimization of cost (ASC) Reliability (LOLE, LOEE)
[58]	✓	✓	✗	✗	✗	✓	NSGA	Minimization of cost (COE) Reliability (LPSP) Power abandonment rate (PAR)
[59]	✓	✓	✗	✗	✗	✓	FPA	Minimization of cost (TNPC) Reliability (LOLE, LOEE)
[60]	✓	✓	✗	✗	✓	✓	BMILP	Minimization of cost (TAC) Minimization of H_2 cost
[61]	✓	✓	✓	✗	✗	✓	GA	Cost (TNPC, COE) and emission (CO_2)
[62]	✓	✓	✗	✗	✗	✓	SOA	Minimization of cost (LCOE)
[63]	✓	✓	✗	✗	✗	✓	IAEO	Minimization of cost (COE) Reliability (LPSP)
[64]	✓	✗	✗	✗	✗	✓	PSO	Minimization of cost (LCOE) and EMS
[65]	✓	✓	✗	✗	✗	✓	WOA	Minimization of cost (TNPC, COE)
[41]	✓	✓	✓	✗	✗	✓	P-Graph	Minimization of cost (COE) and emission (CO_2)
[43]	✓	✓	✗	✗	✗	✓	MILP	Economic and reliability

the island of Stromboli in Italy. The village serves as a typical example of other insular locations found throughout the Mediterranean region. In light of economics and resilience concerns, this research paper [43] proposes a risk-constrained planning strategy for hydrogen-based multi-energy off-grid microgrids using mixed integer linear programming approach. To find the most efficient way to utilize solar and hydrogen resources to provide electrical power to the household while minimizing the impact on the environment, the authors in [44] developed a numerical technique to optimize a solar-hydrogen energy system. An improved bee optimization is paper presents in [45] with a view to enhancing freshwater availability and fulfilling the load demand by modelling six distinct hybrid renewable energy systems (HRES) with and without hydrogen storage technology. The research paper [46] describes the use of an effective metaheuristic method, which is based on artificial bee swarm optimization, to design a hybrid renewable energy system that incorporates photovoltaic, wind, hydrogen, and reverse osmosis desalination technologies. The system is intended to increase the availability of freshwater and satisfy the energy demands of a standalone region in Iran. According to the study conducted by the authors in reference [47], an enhanced optimization algorithm with superior efficacy is proposed for long-term capacity planning and optimization of an off-grid hybrid renewable energy system that comprises wind, fuel cell, and hydrogen storage schemes. The authors suggest three refined adaptations of the global dynamic harmony search approach to perform the pitch adjustment mechanism. The research paper [48] introduces an innovative optimization algorithm that employs the harmony search (HS) method to determine the optimal size of a hydrogen storage system integrated with solar-wind power generation. The primary aim of the optimization problem in question is to minimize the overall system cost while ensuring that the appropriate reliability requirements are met. Several optimization techniques applied for HRES-H_2 are listed in Table 1.

Hybrid power cycle for integrating hydrogen-based renewable energy

Renewable energy sources, such as biomass, fuel cells, biofuels, geothermal, and ocean energy, are employed in combination with conventional energy sources, such as fossil fuels, to create a more sustainable and effective energy system.

For instance, a biomass-fueled power plant can be combined with a gas turbine or steam turbine to create a hybrid power cycle that generates electricity from both renewable and non-renewable sources. Likewise, fuel cells, which convert chemical energy into electrical energy, can be incorporated into power cycles to provide a continuous and reliable source of electricity.

Biofuels, such as ethanol and biodiesel, can also be used in hybrid power cycles to replace fossil fuels and reduce greenhouse gas emissions. Geothermal energy, which exploits the heat from the Earth's interior, can be employed in hybrid power cycles to generate electricity and provide heating and cooling for buildings. In conclusion, ocean energy, which includes technologies such as wave and tidal power, can also be used in hybrid power cycles to generate electricity from renewable sources [66].

Table 2 – Summary of some research on hybrid power cycle for integrating renewable energy.

Ref.	Integrated system type	Objective of design	Main findings
[67]	Photovoltaic-Wind-Biomass-H_2	Optimal dispatch including demand response.	The outcome suggests that the total micro-grid costs have been enhanced by an average of 14.79% and the levelized cost of energy (LCOE) has been increased by 16.211%.
[68]	Photovoltaic-Wind- Hydro-Fuel cell	Techno-economic analysis	Considering the hierarchical control of the system is necessary to guarantee effective functioning and long-term sustainability of the microgrid.
[65]	Photovoltaic-Wind-Tidal-Fuel cell	Reliable designing and economic analysis	Decreasing the net present cost of the hybrid system while meeting the reliability constraints.
[69]	Photovoltaic–Wind-Diesel –Hydrogen–Fuel cell	Triple multi-objective design	Cost and emission reduction.
[70]	Geothermal-Hydro-Biomass-Solar-Wind	Designing	The proposed hybrid energy system produces energy at a lower cost, with increased efficiency and substantial environmental advantages.
[71]	Micro-hydro-Biomass-Biogas-Wind-Photovoltaic	Sizing	The Micro-hydro-Biomass-Biogas-Wind-Photovoltaic has a projected levelized cost of energy (LCOE) of INR 5.47/kWh.

In a nutshell, hybrid power cycles offer a promising solution for integrating renewable energy systems with traditional power generation technologies, providing a more sustainable and effective energy system that reduces greenhouse gas emissions and dependency on fossil fuels. The summary of some research on hybrid power cycle for integrating renewable energy is tabulated in Table 2 below.

Summary and evaluation

Various optimization techniques have been used to optimize the hybrid renewable energy systems with hydrogen (HRES-H_2). The optimization techniques emcompases, the classical, soft computing, and hybrid metaheuristic optimization methods. While classical methods have limitations such as inflexibility and long calculation time, soft computing techniques such as particle swarm optimization (PSO), genetic algorithm (GA), and harmony search (HS) have been utilized to optimize solar and wind power with hydrogen technologies. These optimization techniques have been applied to minimize cost, CO2 emissions, and optimize the size and operational strategies of PV-hydrogen/battery storage systems. Finally, various optimization algorithms have been introduced to determine the optimal size of hydrogen storage systems integrated with solar-wind power generation while minimizing overall system cost and ensuring reliability requirements are met.

Classification of optimization techniques that can be applied to HESS based HRES

There are three categories of size optimization techniques for hybrid renewable energy systems: classical techniques, soft computing techniques, and software tools. Classical techniques use iterative, numerical, analytical, probabilistic, and graphical construction methods, which rely on differential calculus to derive the optimal solution. soft computing techniques, on the other hand, use artificial and hybrid approaches, which can determine the global optimum system and have better convergence and accuracy in finding a set of optimal solutions [72]. Computer software tools represent the third approach to size optimization for HRES, with the most widely used software tool being the Hybrid Optimization Model for Electric Renewables (HOMER) and another tool called Improved Hybrid Optimization by Genetic Algorithm (iHOGA) being used for sizing optimization of standalone HRES. Fig. 3 illustrates the existing size methodologies used for standalone HRES [73].

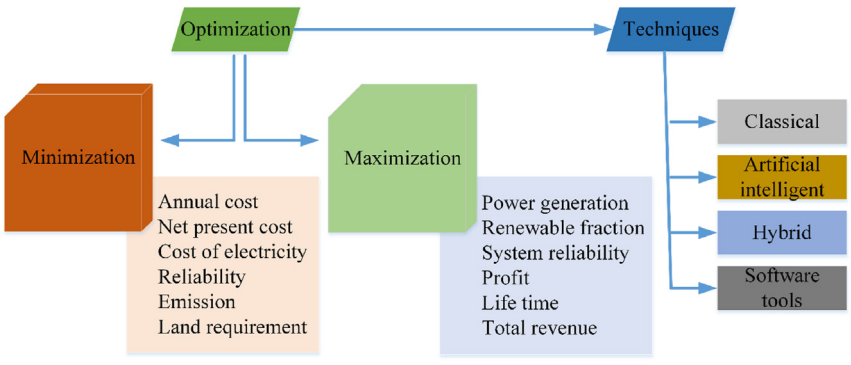


Fig. 3 – Existing size optimization techniques for HRES.

Classical optimization techniques used in HRES

Classical optimization techniques use mathematical formulations to determine global optimal solutions in a deterministic manner. While this approach has the advantage of providing definite answers, the model struggles to handle a large number of variables in a complex space. In the early stages of design and optimization, classical techniques typically use numerical, iterative, analytical, and probability analysis, derived using differential calculus to calculate the most efficient energy models. However, these methods have limited scope when the objective functions are not continuous and/or differentiable. Instead, numerical methods, which determine approximate solutions, are often used in complex optimization problems. Examples of numerical methods applied to optimal sizing of hybrid renewable energy systems include iterative optimization approaches, stochastic and deterministic approaches, and mixed integer linear programming. These methods optimize the size of different components, such as solar PV units, wind generators, fuel cells, electrolyzers, and energy storage units, to attain zero energy deficits, low system cost, and stable power supply. The optimal configuration of the hybrid system varies depending on the location and demand profile. Table 3 shows the benefits and drawbacks of classical techniques.

Recently, there have been a few studies that have used traditional methods to optimize the size of isolated photovoltaic-wind turbine HRES. These studies mostly rely on iterative algorithms. The authors in [79] proposed an optimization methodology based on an iterative technique to optimize the size of an off-grid PV/WT/FC HRES to supply a desalination unit for Kerkennah Island in South Tunisia. Their main objective was to minimize the total capital cost of the system.

Some times ago, a number of studies have utilized linear programming techniques (LP) to optimize the size of an off-grid HRES with PV/WT [80]. [81] utilized a deterministic optimization technique called mixed-integer linear programming (MILP) to find the optimal mix between PV-WT-BS-DG by minimizing LCOE over a lifetime of 20 years. The best system has a renewable component of 90%, according to the findings. The authors of [82] proposed a technique that uses MILP and an exact solve procedure to determine the optimal size and location of the hybrid PV and WT system components, taking into account the energy demand at the consumption points and the energy resource maps. The objective function of the optimization is to minimize the initial system cost, which is also used as a parameter to assess the system. The study concluded that the optimal location, in addition to optimal size, helps reduce the initial investment costs.

Several studies have suggested using graphical techniques to analyze various aspects of off-grid hybrid systems. For instance, the authors in [83] proposed a graphical technique for decaying CO_2 emissions in multi-product plants, and assessed the technological and economic feasibility. Another study in [84] developed a graphical user interface model using time series analysis to optimize off-grid hybrid systems based on minimizing the life cycle cost and balancing energy storage with diesel fuel generator usage. Another authors in [85] study examined the sizing of PV systems capable of delivering continuous power to loads using a geometrical construction method based on long-term solar radiation data.

Some studies have proposed probabilistic methods that rely on probability theory for decision-making, but this may not ensure the optimal solution. Another researchers in [86] introduced a new probabilistic method that estimates the long-term performance of a hybrid solar-wind power system. Another study [87] developed a hybrid programming model that integrates probabilistic analysis with artificial neural network training and a genetic algorithm optimization model to analyze the behavior of a hybrid solar-wind energy storage system under unpredictable environmental conditions. The summary of some HRES optimized using analytical method is presented in Table 4.

Summary and evaluation

Classical optimization techniques use mathematical interpretations to determine global optimal results in a deterministic approach, but struggle to handle complex spaces

Table 3 – Advantages and drawbacks of classical optimization techniques.

Technique	Advantage	Drawback	Ref.
Analytical	Calculating the size is much easier compared to MCS, and requires less computing power and time.	Can be challenging to estimate the coefficients of mathematical equations that are dependent on location, which in turn requires longer computation time and limits flexibility in designing systems.	[74]
Iterative	Simple to write the code and has the ability to detect threats in the early stages.	The approach does not optimize the area and slope angle of PV panels, the swept area of wind turbines, or installation heights, resulting in suboptimal solutions and increased computational effort.	[75]
Graphical	Straightforward understanding	Graphical methods usually optimize only two parameters at a time, such as photovoltaic and battery or photovoltaic and wind turbine.	[76]
Numerical	There is no defined method for obtaining a sequence of approximate solutions.	As the computation includes linear shifts in choice variables, hence, results in minimal solutions.	[77]
Probabilistic	The easiest sizing methods do not require time-series data.	These methods generate results that are not ideal for identifying the best solution, because one or two performance indicators are considered when optimizing.	[78]

Table 4 – Summary of available HRES optimized using classical method.

Ref.	Configuration								Techniques	Objective function
	PV	WT	MHT	BIO	D_{gen}	Others	BAT	HESS		
[88]	✓	✓	✗	✗	✗	✗	✗	✓	Analytical	Minimize Cost
[89]	✓	✗	✗	✗	✗	✗		✗	Analytical	Optimize of payback period
[90]	✓	✗	✗	✓	✓	✗	✓	✗	Iterative	Minimize Cost
[41]	✓	✓	✓	✗	✗	✗	✗	✗	Graphical	Cost and emission
[91]	✓	✓	✗	✗	✗	✗	✓	✗	Probalistic	Reliability
[92]	✓	✓	✗	✗	✗	✗	✗	✗	Probalistic	Minimize Cost
[93]	✓	✓	✗	✗	✓	✗	✓	✗	Probalistic	Net present value
[94]	✓	✗	✗	✗	✗	✗	✗	✗	Stochastic	Cost minimization
[95]	✓	✗	✗	✗	✗	✗	✓	✓	MILP	Minimize cost
[96]	✓	✗	✗	✗	✗	✗	✓	✓	MILP	Cost and emission
[97]	✓	✗	✗	✗	✗	✗	✓	✗	Stochastic	Riliability Life cycle cost
[98]	✓	✗	✗	✗	✗	✗	✗	✗	Numerical	Life cycle cost
[99]	✓	✓	✗	✗	✗	✓	✓	✗	Analytical	Minimize Cost

with a large number of variables. Numerical techniques, which determine estimated solutions, are often used instead. Examples of numerical techniques used to optimize the size of hybrid renewable energy systems include iterative optimization methods, stochastic and deterministic approaches, and mixed integer linear programming. Graphical techniques and probabilistic methods have also been proposed. Studies have optimized the size of off-grid PV/WT/FC HRES to supply energy demands, with objectives including minimizing total capital cost, minimizing LCOE over a lifetime of 20 years, and minimizing initial system cost. These optimization techniques can vary the optimal configuration of the hybrid system depending on location and demand profile.

Artificial intelligence techniques

The HRES design is complicated because of the uncertainties related to renewable resources and technical factors, as well as the constraints imposed by the site location and system components. Traditional methods are not effective in addressing such intricate issues. As a result, in the past ten years, soft computing techniques that rely on meta-heuristic algorithms have been widely employed [72]. There are two types of sizing optimization methodologies: single objective optimization (SOO) and multi-objective optimization (MOO). SOO aims to find the best possible solution by minimizing or maximizing a single defined value. On the other hand, MOO utilizes multiple objective functions to generate a range of possible solutions that trade-off different objectives, allowing decision-makers to choose the most appropriate one based on the specific problem conditions. Both SOO and MOO can achieve efficient results by identifying global optimal solutions in the Pareto set, depending on the objectives' scope. This leads to improved cost-effectiveness and reliability of the hybrid renewable energy system (HRES) combination [100].

Typically, traditional approaches rely on single algorithms that use SOO function. In contrast, modern methods use both single and hybrid algorithms to tackle both SOO and MOO problems. This greater flexibility enables soft computing techniques to handle more complicated optimization problems and produce more precise outcomes.

Single optimization algorithm applied on HRES

The subsequent sub-sections will review the application of single algorithms, encompassing both classical and artificial techniques, in resolving size optimization problems in HRES.

Researchers use various types of algorithms to determine the optimal sizing for hybrid systems, such as GA, DE, NSGA, MBA, PSO, multi-objective line-up competition algorithms, ACO, preference-inspired coevolutionary algorithms, BBO, CS, discrete harmony search, SA, ABSA, IFFA, BFA, ANN, fuzzy logic and many more others [101]. Table 5 provides a synopsis of each technique.

Hybrid artificial intelligence techniques applied to HRES

Single-optimization algorithms that are artificially designed can quickly and accurately generate optimal solutions while requiring less time for convergence. However, as the use of HRES becomes more widespread, particularly in remote areas and islands, there is a growing need for even more advanced and precise optimization methods. As a result, hybrid algorithms have emerged as a popular choice for sizing optimization in off-grid HRES. These hybrid algorithms combine two or more single algorithms, whether modern or classical, to take advantage of their unique strengths and address complex optimization problems that involve various linear and nonlinear constraints.

The authors in [122] developed sizing model for a HRES that includes PV, WT, D_{gen}, and battery components. The model uses a hybrid artificial intelligence algorithm called enhanced differential evolution and particle swarm optimization (DE and PSO) to find the optimal sizing for each component. The model takes into account different optimization objectives and provides the designer with several options to choose from based on their preferences. In reference [123], the authors have created a hybrid algorithm called the water cycle algorithm and moth-flame optimization. This algorithm is used to determine the optimal design of an off-grid hybrid solar PV, biogas generator, pumped hydro energy storage, and battery

Table 5 – A summary of single optimization techniques applied on HRES.

Techniques	Configuration					Storage		Essential formulation	Ref.
	PV	WT	MHT	BIO	D_{gen}	BESS	HESS		
GA	✓	✓	✗	✗	✗	✓	✗	$DPSP = \frac{\sum_{i+1}^{h} DPS_i}{\sum_{i+1}^{h} demand_i} \times 100$	[102]
DE	✓	✓	✓	✓	✗	✓		$NPC_T = In_C + O\&M_C + R_C + F_C$	[103]
PSO	✓	✓	✓	✗	✗	✓		$LPSP = \frac{\sum_{t=1}^{r} LPS^t}{\sum_{t=1}^{r} E_{Load}t}, LPSP \leq LPS^{max}$	[104]
FA	✓	✓	✗	✗	✗	✓	✗	$Minimize\ EC = (''\$''/kWh) = \frac{LCC \times CFR}{\sum_{t-1}^{8760} E_{Load}(t)}$	[105]
								$LDR = \frac{\sum_{t-1}^{T}(E_{Load}(t) - E_{Surp}(t)}{\sum_{t-1}^{T} E_{Load}(t)}\%$	
IPSO	✓	✓	✗	✗	✓	✓	✗	$LCOE = \frac{C^I + C^A \sum_{T=1}^{N} d^T}{E^A \sum_{T=1}^{N} d^T}$	[106]
IACO	✓	✓	✗	✗	✓	✓	✗	$LPSP = \frac{\sum Time(if P_{avail}(t) < P_L(t))}{T}$	[107]
GA	✓	✗	✗	✓	✗	✓	✓	$LCOE = \frac{TAC}{P_{L,annua}}$	[108]
								$LPSP = \sum_{t=1}^{8760} \left(\frac{P_L(t) - P_{gen}(t)}{P_L(t)}\right)$	
MFOA	✓	✓	✗	✗	✗	✓	✗	$ELF_{Load} = \frac{1}{n}\sum_{t=1}^{n} \frac{Q_{Load}(t)}{P_{Load}(t)}$	[109]
ABC	✓	✓	✗	✗	✓	✓	✗	$LCOE = \frac{ASC(''\$''/year)}{Total\ energy\ served(kWh/year)}$	[110]
HS	✓	✓	✗	✗	✓	✓	✓	$C_T = C_{Cpt} + C_{Mtn} + C_{Fuel}$	[111]
MILP	✓	✓	✗	✗	✗	✓	✓	$Cost = m(\sum(P_G(t) \times \$/kWh(t) + B_{Dc}(t)))$	[112]
PSO	✓	✓	✗	✗	✓	✓	✗	$COE\left(\$/kWh\right) = \frac{TNPC}{\sum_{n=1}^{8760} P_h(n) \times C_{pv}}$	[113]
GWOA	✓	✓	✗	✗	✓	✓	✗	$T_{npc} = C_C + O\&M_C + R_C + F_C$	[114]
								$COE = \frac{TotalNPC}{\sum_{n=1}^{8760} P_h(n) \times CRF}$	
GOA	✓	✓	✗	✗	✓	✓	✗	$LCOE = \sum_{n=1}^{N} \frac{Cost_{System}/(1+r)^n}{Energy_{System}/(1+r)^n} \left(\frac{\$}{kWh}\right)$	[115]
EMPA	✓	✓	✗	✗	✓	✓	✗	$mEmission = \sum_{i=1}^{n} (E_{CO2} + E_{SO2} + E_{nox}) P_{Out_i}$	[116]
NSGA-II	✓	✓	✗	✓	✗	✗	✓	$MinCOE = \frac{NPC \times CRF(i, L)}{\sum_{t=1}^{8760} E_S}$	[117]
SSA	✓	✓	✗	✗	✓	✓	✗	$LPSP = \frac{\sum(P_{Load} - P_{pv} - P_{wt} + P_{SOC_min} + P_{die})}{P_{Load}}$	[118]
PSO	✓	✓	✗	✗	✓	✓	✓	$LCOE = \frac{C_{NPC,tot}}{\sum_{j-1}^{n} \frac{E_{tot,j}}{(1+d)^j}}$	[119]
ALO	✓	✓	✗	✗	✗	✓	✗	$COE = \frac{(CRF^* \sum_x NPC_x) + C_{grid} - R_{grid}}{E_{served} + E_{grid-selling}}$	[120]
GOA	✓	✓	✗	✗	✓	✓	✗	$COE = \frac{TNPC}{\sum_{t=1}^{8760} P_L(t)}$	[121]

system for a radio transmitter station in India with a view to achieve optimization both technological and economic benefits. The study described in reference [124] makes use of three different algorithms, namely Harmony Search (HS), Jaya, and Particle Swarm Optimization (PSO), to determine the optimal design for a hybrid renewable energy system (HRES) that includes Wind, Photovoltaic, Biomass, and Battery technologies. The main objective of this study is to meet the electricity demands of customers in a cost-effective, efficient, and reliable manner. In their study [125], utilized a combination of a genetic algorithm (GA) and an exhaustive search technique to determine the optimal sizing of a standalone hybrid PV, WT, and battery system while taking into account socio-demographic factors. The optimization objective was to minimize total cost (TC) and ensure system reliability by meeting load demand. The reliability parameter used in the optimization was LPSP, while socio-demographic factors were used as a social assessment parameter. The study investigated the impact of socio-demographic factors on the sizing of hybrid energy systems (HES) by examining the energy usage patterns of six different electrical users and their influence on the HES size. The authors discovered that the optimal size of one user, achieving 0% LPSP, cannot be precisely the same as that of other users. In such cases, the system's generation and storage capacity should be increased to meet all user load requirements, resulting in higher system costs. The combination of PSO and SA has been used to enhance the collective behavior of a group and the effectiveness has been assessed using various benchmark functions to achieve global minimization [126]. In reference [127], authors used an

Table 6 – Summary of some hybrid optimization techniques applied on HRES.

Techniques	Configuration					Storage		Essential formulation	Ref.
	PV	WT	MHT	BIO	D_{gen}	BESS	HESS		
EDE, PSO	✓	✓	✗	✗	✓	✓	✗	$LCOE = \frac{NPC}{\sum_{h=1}^{8784} P_{Load}} \times CRF$	[122]
WCA, MOA	✓	✗	✗	✓	✗	✓		$LLP = \frac{\sum_{t=1}^{T} DP(t)}{\sum_{t=1}^{T} E_{actualload}(t) \times \Delta t}$	[123]
HS, Jaya, PSO	✓	✓	✗	✓	✗	✓		$C_{NPC} = \sum_x C_I + C_R + C_{O\&M} + C_F - S$	[124]
CS, HS, SA	✓	✓	✗	✗	✗	✗	✓	$LPSP = \frac{\sum_{t-1}^{T} LPS(t)}{\sum_{t-1}^{T} E_{Load}(t)}$	[128]
FA, HSO	✓	✓	✗	✗	✗	✗	✓	$LPSP = \frac{\sum_{t=1}^{T} P_{deficit}(t).\Delta t}{\sum_{t=1}^{T} P_{demand}(t).\Delta t}$	[129]
BBO, PSO	✓	✓	✗	✗	✗	✓	✗	$Total\ cost = \sum_{i=s,w,b} (I_i - S_{P_i} + OM_{P_i}) + C_{grid}$	[130]
HBB–BC	✓	✓	✗	✗	✗	✓	✗	$Min\ TPC = C_{INV} + C_{MAIN} + C_{REP}$	[131]
PSO, WCT	✓	✓	✗	✗	✗	✓	✗	$COE = \frac{NPC}{\sum_{h=1}^{8784} P_{Load}} CRF$ $LPSP = \frac{\sum_{t=1}^{T} LPS(t)}{\sum_{t=1}^{T} P_{Load}(t)}$	[132]
MCOA, PSO	✓	✓	✗	✗	✗	✓	✗	$TAC = ACC + AMC$	[133]
HS, HS-CS	✓	✓	✗	✗	✗	✓	✗	$Minimize\ TLCC(A_{PV}, A_{WT}, N_{BAT})$ $LPSP = \frac{\sum_{t=1}^{T} LPS(t)}{\sum_{t=1}^{T} P_{Load}(t)}$	[134]
HGWO, SCA	✓	✓	✗	✗	✗	✗	✓	$LIP = \frac{\sum_{t-1}^{T} [P_{Load}(t) - P_{Inv-Load}(t)]}{\sum_{t-1}^{T} [P_{Load}(t)]}$	[135]

optimization approach to determine the optimal size of an off-grid HRES for Kuala Terengganu in Malaysia. The authors hybridized an iterative algorithm and GA to optimally size the proposed HRES. The size optimization was performed at different LPP values based on the available daily wind speed and solar radiation. Table 6 present the summary of some hybrid optimization techniques applied on HRES.

Summary and evaluation

Designing hybrid renewable energy systems (HRES) can be complicated due to uncertainties and constraints. Traditional methods are not effective in addressing these issues, so soft computing techniques that use meta-heuristic algorithms have been widely employed in the past decade. There are two types of sizing optimization methodologies: single objective optimization (SOO) and multi-objective optimization (MOO). Soft computing techniques use both single and hybrid algorithms to tackle SOO and MOO problems, providing greater flexibility and more precise outcomes.

Software tools used for HRES analysis

Having a thorough comprehension of the hybrid system models and software tools that are available for research purposes, including their characteristics, limitations, and the preferences of users, is crucial. This section provides an in-depth examination of the principal features of the significant software programs created for hybrid system design, as well as a comparative evaluation.

The software tools for HRES are classified into four types: pre-viability, sizing, simulation, and open architecture investigation tools. Pre-viability tools, such as RETScreen, are typically used for estimating system size and performing financial analyses. The sizing tools determine the optimal size for each system component and provide information about energy flows between components, such as HOMER. Simulation tools, such as HYBRID 2, require the user to specify the details of the components in order to obtain a comprehensive view of the system's behaviour. The following discussion focuses on software tools that are identified for effectively using them in various research studies, particularly in hybrid systems with renewable energy components. The aim is to provide a fundamental understanding of how these tools can be utilized.

Furthermore, TRNSYS software is used to evaluate how sophisticated control methods impact the energy and financial effectiveness of a home heat pump system that comes with a PV system and storage device, which functions in various temperature ranges [136]. This analysis aims to optimize the performance of the system in terms of energy and economic efficiency. In reference [137], the authors used HOMER software to examined the renewable energy resources that were accessible in the region and assessed the economic, technical, and environmental factors of five different energy sources: diesel system, photovoltaic with storage system, hybrid photovoltaic/diesel with and without storage systems, and hybrid photovoltaic/wind with storage system.

The article [138], provides a study on the technical and economic aspects of a hybrid photovoltaic/diesel and battery system configuration for five residential homes located in Barhim Quarters, Katsina, Nigeria. The analysis was conducted using the HOMER optimization software to optimize the system's technical and economic performance. The iHOGA software was utilized to conduct single and multi-

objective optimization of a HRES for a typical household in France. Multi-objective optimization aimed to reduce CO2 emissions and unmet load whilst keeping a little higher NPC compared to the single goal of minimizing the NPC. In another application, the iHOGA tool was used to optimize a HRES for the residents of Mucura island, demonstrating the feasibility of reducing operating costs by decreasing dependence on diesel fuel, optimizing the use of local renewable energy sources, and ensuring energy availability throughout the year. The results indicate the potential for sustainable and cost-effective energy solutions [139]. The RETScreen software was employed to determine the most suitable size for renewable energy systems for two different problem scenarios. The optimal sizing was evaluated based on various objective functions, considering both scenarios with and without renewable energy system subsidies [140]. Reference [141] reports the investigation of a 200 KW solar power system at VIT University using TRNSYS simulation tool. Table 8 provides a summary of various studies conducted on HRES for different locations using the mentioned software tools. The article [142] explores four different scenarios related to the use of PV systems in residential areas, in response to new legislation. RETScreen software was used to examine each scenario, along with a brief analysis of their economic and environmental impacts. The fourth scenario, proposed in the article, aims to increase the adoption of PV systems in households, and the impact on the economy was studied, specifically in terms of government incentives for installing these systems. Table 7 provide various summary of various studies conducted on HRES for different locations using software tools.

Summary and evaluation

The section examines four categories of software tools for hybrid renewable energy systems (HRES): pre-viability, sizing, simulation, and open architecture investigation tools. A review of software tools that can be used in HRES research studies, particularly for hybrid systems with renewable energy components is also provided. The article includes case studies that utilize various software tools such as HOMER, RETScreen, TRNSYS, and iHOGA to optimize technical and economic performance of HRES. The authors also present a summary of various studies conducted on HRES for different locations using the mentioned software tools in a tabular form.

Energy management strategy

Electricity generated from renewable energy systems is not reliable and fluctuate with time. To tackle this problem, a number of design criteria must be considered. One way is by combining two or more RESs with energy storage devices to achieve a hybrid RES that can provide backup power. The hybrid RES also requires strategic control through an Energy Management System (EMS) to coordinate the power flow of the RESs that made up the hybrid system. Depending on the

Table 7 – Summary of various studies conducted on HRES for different locations using software tools.

Software	System configuration							Location	Highlight of the outcome	Ref.
	PV	WT	MHT	BIO	D_{gen}	Others	BESS			
HOMER	✓	✗	✓	✗	✓	✗	✓	Iraq	The analysis of certain parameters that are not commonly reported, battery efficiency, capital cost of photovoltaic multiplier etc., showed that changes in these factors have a noteworthy impact on the power flow and economic evaluation of the system.	[143]
HOMER	✓	✓	✗	✗	✓	✗	✓	Australia	The NPC, LCOE, and CO_2 emissions are greater for the diesel-only system in comparison to the optimized HRES.	[144]
HOMER Pro.	✓	✓	✗	✗	✗	✗	✓	Egypt	The total costs of energy and LC for the system are $751,597 and $0.2374 per kWh, correspondingly.	[145]
iHOGA	✓	✓	✗	✗	✓	✗	✓	Paris	While not as cost-effective as using the grid, such an HRES can promote decarbonization in development, allowing France to make a significant contribution to COP21 obligations.	[146]
iHOGA DIgSILENT	✗	✓	✗	✗	✓	✗	✓	Malaysia	The results of the simulation, which include the NPC, COE, and CO_2, show an annual reduction of approximately 72%, 73.5%, and 82% respectively, in comparison to traditional power plant.	[147]
HYBRID2	✓	✓	✗	✗	✗	✓	✓	Chicago	The results shows that the RE available at the site is enough to fulfill the load demands. Fuel cells is such locations.	[148]
TRNSYS	✓	✗	✗	✗	✗	✓	✗	Cyprus	Lowering the initial and operational expenses makes the HRES system financially feasible.	[149]
RETScreen	✓	✓	✗	✗	✓	✗	✓	China	The HRE generation is determined to be 82.1% and 16.2%, respectively, and there is a projected drop in annual CO_2.	[150]

Table 8 – Summary of the control procedure, optimization purposes, limits, and EMS category from the referred literature.

Optimization goals	System configuration							Control algorithm	Design limitations	EMS highlight	Ref.
	PV	WT	MHT	BIO	D_{gen}	HESS	BESS				
Fulfill load demand	✓	✓	✗	✗	✓	✓	✓	Flow Chart	SoC, H_2 volume, power balance	The primary focus is on using BES to provide electricity when there is a shortage and store any excess energy. FC and electrolyzer will only be used based on the minimum and maximum SoC of the battery.	[157]
	✓	✓	✗	✗	✓	✓	✓	MPC	SoC, H_2 volume, power balance	If there is an excess of energy, the priority is given EL to create hydrogen, followed by charging the BES. If there is a shortfall in energy generation, the stored energy in the BES and HT will be checked, and once again, the use of hydrogen will be prioritized.	[158]
	✓	✓	✗	✗	✓	✓	✗	Matlab simulink	H_2 reserve, power balance	In this plan, energy demand is met by the power generated from PV and WT. If there is excess energy, the system will supply EL to produce hydrogen, and if there is a shortage of energy, the FC will compensate for the shortage.	[159]
	✓	✓	✗	✗	✓	✓	✗	Flow chart PSO	Power balance	Electrolyzer and fuel cell compensate for any excess or shortfall in power generation.	[160]

objectives of the design, the EMS of the hybrid RES aims to ensure a consistent supply of electricity to the load, expand system performance, exploit resource utilization, minimize system operational costs, and extend the lifetime of the system [151].

In the context of energy hub concept, there are several methods that can be used to manage and optimize the energy flows within the system. These include.

- Rule-based method: This method involves the use of predefined rules to manage the energy flows within the system. For instance, the system may prioritize the use of renewable energy sources over fossil fuels through certain times of the day, or may cap the energy consumption of certain devices during peak demand periods.
- Optimization-based method: This method involves the use of mathematical optimization models to optimize the energy flows within the system. These models take into account various factors such as energy demand, availability of energy sources, and the cost of energy to determine the optimal energy flow pattern that minimizes energy costs or maximizes energy efficiency.
- Learning-based method: This approach involves the use of machine learning algorithms to learn and adapt to the changing energy demand and supply models within the system. For instance, the system may use clustering algorithms to identify patterns in energy utilization and amend the energy flow appropriately.
- Reinforcement learning (RL) based approach: This approach involves the use of RL algorithms to optimize the energy flow within the system based on feedback from the system. RL algorithms learn from the feedback provided by the system to identify the optimal energy flow pattern that maximizes a particular reward function, such as energy efficiency or cost savings.

Each of these approaches has advantages and drawbacks and may be more suitable for different types of energy hub systems. The choice of method will be determined by several factors such as the size of the system, the complexity of the energy flows, and the specific aims and objectives of the energy hub. The use of machine learning techniques has transformed the way modern energy systems are monitored, controlled, and optimized, including both conventional fossil fuel power plants and renewable energy systems [152–154].

Finding an accurate load profile is another critical component of the optimization problem. Accurate load profiles are necessary for optimization of energy management and policy recommendations, as policy-makers seek to balance environmental compatibility, economic efficiency, and security of supply [155]. Data analysis tools and statistical methods have been reported in the literature for constructing the load profiles when load profiles of the case study region is not available. The techniques requires analyzing historical consumption data and using statistical techniques to identify models and trends in energy use. In addition, clustering or regression analysis have also been used to construct load profiles using available climatological data. In some cases, load profiles can also be estimated based on the energy requirements of individual appliances or devices. Additionally, load forecasting methods can be used to predict future load profiles based on historical data, weather patterns, and other factors [156].

The majority of previous research on this topic has focused on simulating strategies to meet load demand without taking

into account economic and technical optimization criteria, such as the degradation of hydrogen (H_2) equipment. In the following section, this study provides a review of various hydrogen technology based Energy Management Systems (EMS) that have been used in scientific literature, explaining the goals of each strategy.

Strategy that consider demand fulfillment

Demand satisfaction is a crucial link between load and distributed resources, making energy management of micro-grids more practical. While demand satisfaction is an essential factor to consider for micro-grid energy management, others factors also need to be considered. An operational optimization strategy is necessary to achieve efficient and cost-effective micro-grid operation. The main goal of this strategy is to meet demand, and as a result, the control algorithm focuses on three design criteria: hydrogen reserve, battery state of charge, and power balance. The decision variables (H_2 reserve, BT SoC, and power balance) establish the operating limits for the battery, fuel cell, and electrolyzer. The strategy is recognized for by the ease of control and design, and the operation is presided by straightforward flowchart algorithms. The application of sizing that implements this strategy is also make simpler. However, the solution attained from this approach is suboptimal from both a technical and economic viewpoint due to the absence of optimization parameters based on the operating costs of the system and equipment degradation. Table 8 presents a review of the control procedure, optimization purposes, limits, and EMS category from the referred literature.

Strategies that consider technical and economic decision issue

This approach aims to enhance the performance of the system and guarantee that demand is met. Combine technical and economic factors are taken into account in this plan to decrease maintenance expenses and extend the lifespan of the equipment. From both an economic and technical standpoint, the solutions offered by this approach are advantageous because the method incorporates equipment deterioration and cost parameters into a multi-objective function. The solution generated establishes the appropriate power supplied by each component in the system, which guarantees optimal performance and power balance. The primary benefit of this plan is the ability to provide optimal system response from both technical and economic angles. System performing constraints and lifespan are taken into account when creating the optimization function. However, the optimization algorithms used are complicated and may increase the system's complication in real-world applications. Table 9 presents a summary of the control algorithm, optimization goals, constraints, and EMS description obtained from the reviewed literature.

Table 9 – Summary of the control algorithm, optimization goals, constraints, and EMS description obtained from the reviewed literature.

Optimization goals	System configur					Control algorithm	Design limitations	EMS highlight	Ref.
	PV	WT	D_{gen}	HESS	BESS				
Minimize cost Fulfill demand Extend life expectancy	✓	✓	✗	✓	✗	Flow Chart	SoC, H_2 volume, power balance	The primary goal of this optimization approach is to reduce the overall cost of the system while maintaining a particular level of reliability (either 1%, 2%, or 3%).	[161]
	✓	✓	✗	✓	✓	Flowchart with linear programing	SoC, H_2 volume, balance of power, degradation, and cost.	The results of this optimization process indicate the reference power for the BES, FC, and EL, given the hydrogen reserve SoC of the BES.	[162]
	✓	✓	✓	✓	✓	PSO	SoC and power balance	There is no predetermined preference for the use of specific system elements; rather, the priority is determined by the optimization algorithm's outcome.	[163]
	✓	✓	✗	✓	✓	Flow chart Adaptive PSO	SoC, cost reduction, power balance, and H_2 volume	The system provided better results in terms of reliability and per unit cost when all design parameters such as the number of PV panels, WT, BES, FC, H_2 volume, and depth of discharge of the BES were considered.	[164]
	✓	✓	✗	✓	✓	Flow chart	SoC, cost reduction, power balance, and H_2 volume	If the energy resources in the storage devices are equal, a techno-economic factor is employed to determine which storage device supplies or absorbs energy to maintain a power balance in the system.	[165]

Summary and evaluation

An Energy Management System (EMS) is required to coordinate the flow of power between different components. Rule-based, optimization-based, learning-based, and reinforcement learning (RL) based approaches are used to optimize hybrid energy systems for energy efficiency. Rule-based energy management strategy is prevalent in practical applications, while optimization-based strategies involve solving complex optimization problems using algorithms. RL-based energy management strategies use machine learning techniques to train a controller to make decisions based on the current state of the system and a set of predefined goals. The decision variables establish the operating limits for the battery, fuel cell, and electrolyzer. The optimization function used considers technical and economic factors to cut maintenance costs and extend the lifespan of the equipment. The main benefit of this plan is the ability to give optimal system response from both technical and economic points, but the optimization algorithms used are complicated and may increase the system's complication in real-world applications.

Major findings

Hybrid renewable energy systems (HRES) that use hydrogen (HRES-H2) can be optimized using classical, modern, and hybrid metaheuristic optimization techniques. However, classical methods have limitations such as inflexibility and long calculation time, while soft computing techniques such as particle swarm optimization (PSO), genetic algorithm (GA), and harmony search (HS) have been employed to optimize solar and wind power with hydrogen technologies. These techniques aim to minimize costs, reduce CO2 emissions, and optimize the size and operational strategies of PV-hydrogen/battery storage systems. Different optimization algorithms, including iterative optimization approaches, stochastic and deterministic approaches, mixed integer linear programming, graphical techniques, and probabilistic methods, have been used to determine the optimal size of hydrogen storage systems integrated with solar-wind power generation.

Soft computing techniques that use meta-heuristic algorithms have been widely employed to address the complexities of designing HRES due to uncertainties and constraints. Single objective optimization (SOO) and multi-objective optimization (MOO) methodologies have been used, with MOO preferred due to the ability to identify global optimum Pareto-set solutions. Software tools for HRES can be divided into four categories: pre-viability, sizing, simulation, and open architecture investigation tools. Various software tools, such as HOMER, RETScreen, TRNSYS, and iHOGA, have been utilized to optimize technical and economic performance of HRES in different case studies. The authors provide a summary of different studies conducted on HRES for various locations using the mentioned software tools in a tabular form.

To coordinate the flow of power between different components, an Energy Management System (EMS) is necessary. Rule-based, optimization-based, learning-based, and reinforcement learning (RL) based approaches are used to optimize hybrid energy systems for energy efficiency. RL-based energy management strategies use machine learning techniques to train a controller to make decisions based on the current state of the system and a set of predefined goals. The optimization function used considers technical and economic factors to decrease maintenance expenses and extend the lifespan of the equipment. The primary benefit of this plan is the ability to provide optimal system response from both technical and economic angles, but the optimization algorithms used are complicated and may increase the system's complexity in real-world applications.

Conclusion

This study primarily dealt with classical techniques, artificial intelligence-based optimization methods, hybrid algorithms, and commercial software tools used for the optimal sizing and energy management of hybrid renewable energy systems with emphasis on hydrogen storage based renewable energy system. Although most of these optimization techniques demonstrate high performance and the ability to handle complex issues, often, the weakness of these solutions is complexity and sensitivity to specific problems. Classical techniques, while the most useful strategy for economic optimization, suffer from limitations in the parameter space available for optimization. Classical methods are associated with several drawbacks, such as rigidity, difficulty in locating a global optimum, and the inability to adapt to dynamic changes. In contrast to classical methods, artificial intelligence algorithms can provide a solution that is globally optimal and has a fast convergence speed. In the context of technical, economic, and performance comparisons, These algorithms have been observed to be superior in solving multiobjective optimization problems. These techniques are widely used in optimization because of the higher efficiency, greater accuracy, and fast convergence rates. However, artificial intelligence methods are more complex and require a delicate balance between the exploration and exploitation phases. Additionally, their implementation necessitates significant hardware performance due to the complicated processes and codes involved. Nonetheless, these techniques provide speedy and precise results while maintaining high efficiency. Combining two or more algorithms can also enhance problem-solving efficacy and reliability. Among the many software tools available, HOMER is the most commonly used software due to the ability of allowing for the highest level of renewable energy combination and allows for optimization and sensitivity analyses, resulting in a faster and more straightforward evaluation of numerous potential system configurations. However, a disadvantage is that only a single objective function is employed. Scientists are currently focusing on artificial intelligence methods. Because of the ability to search for global optima, execute high-precision computations, and converge quickly, such techniques are seen to be superior to traditional methods. Despite these advantages, artificial approaches face challenges due to the greater number of inputs required. Studies have shown that hybrid algorithms are more popular than single algorithms for delivering more precise outcomes and effectively addressing multi-objective optimization problems. Moving forward, researchers should prioritize

comprehensive investigations to develop efficient artificial intelligence methods, crossbreed them to be able to proficiently handle diverse objective functions.

Declaration of competing interest

The authors declare that they have no known competing financial interest or personal relationships that could have appeared to Influence the work reported in this paper.

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